

Hello there, my name is Nikolay Vetsov and I am doing projects as Data Analyst Intern.

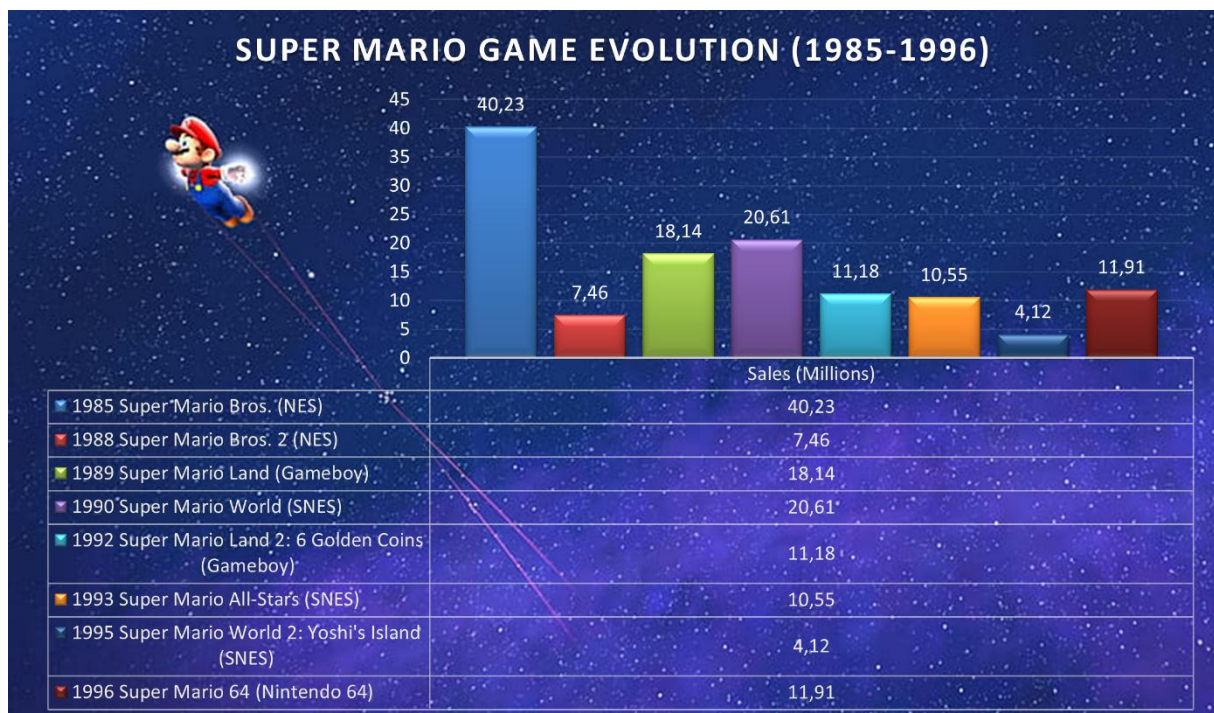
This will be my project №15. I will show you some unique visualizations (mostly by Excel), which are attractive enough, that the person with the interest cannot refuse to see the analysis so easily.

I will be showing you mini datasets, visualizations and recommendations. So, let's not waste more time and start.

Go to the next page:

VISUALIZATION 1 – Super Mario Evolution

Year	Game	Platform	Sales (Millions)
1985	Super Mario Bros.	(NES)	40,23
1988	Super Mario Bros. 2	(NES)	7,46
1989	Super Mario Land	(Gameboy)	18,14
1990	Super Mario World	(SNES)	20,61
1992	Super Mario Land 2: 6 Golden Coins	(Gameboy)	11,18
1993	Super Mario All-Stars	(SNES)	10,55
1995	Super Mario World 2: Yoshi's Island	(SNES)	4,12
1996	Super Mario 64	(Nintendo 64)	11,91



(Created using Microsoft Excel)

Data source: https://en.wikipedia.org/wiki/Super_Mario

Data source 2: <http://vgsales.fandom.com/wiki/Yoshi>

This is a visualization which shows the evolution of the Super Mario games – different versions, platforms and the revenue as well. As you can see here it is not each new game to generate bigger profit than the older one.

My recommendation: If you are planning to win more and more with each new game released, I can recommend you the following:

- *Watch on the reviews of the games by your users. They will tell you what they like in the game what they do not like and what do they prefer;*

- *Each new game must be more realistic and even more effective in 3D. Game speed is so good – it should be not too slow and not too fast. So, experimenting before game release is a great idea;*
- *You can hire someone (secret agent) to ask questions online. For example like: “How do you imagine the next Super Mario Game?”;*
- *Improving the Super Mario game music, make a changes for the new version;*
- *Adding new power in the game (different than the mushroom, flower, star). But do not abuse. Better be just 1 new power for the new game;*
- *Often have new ideas, new scenarios – that may help you for a better new games and therefore for bigger profit;*
- *And of course, you can watch the competitor’s ideas and their review from the players, which will surely help you for better ways.*

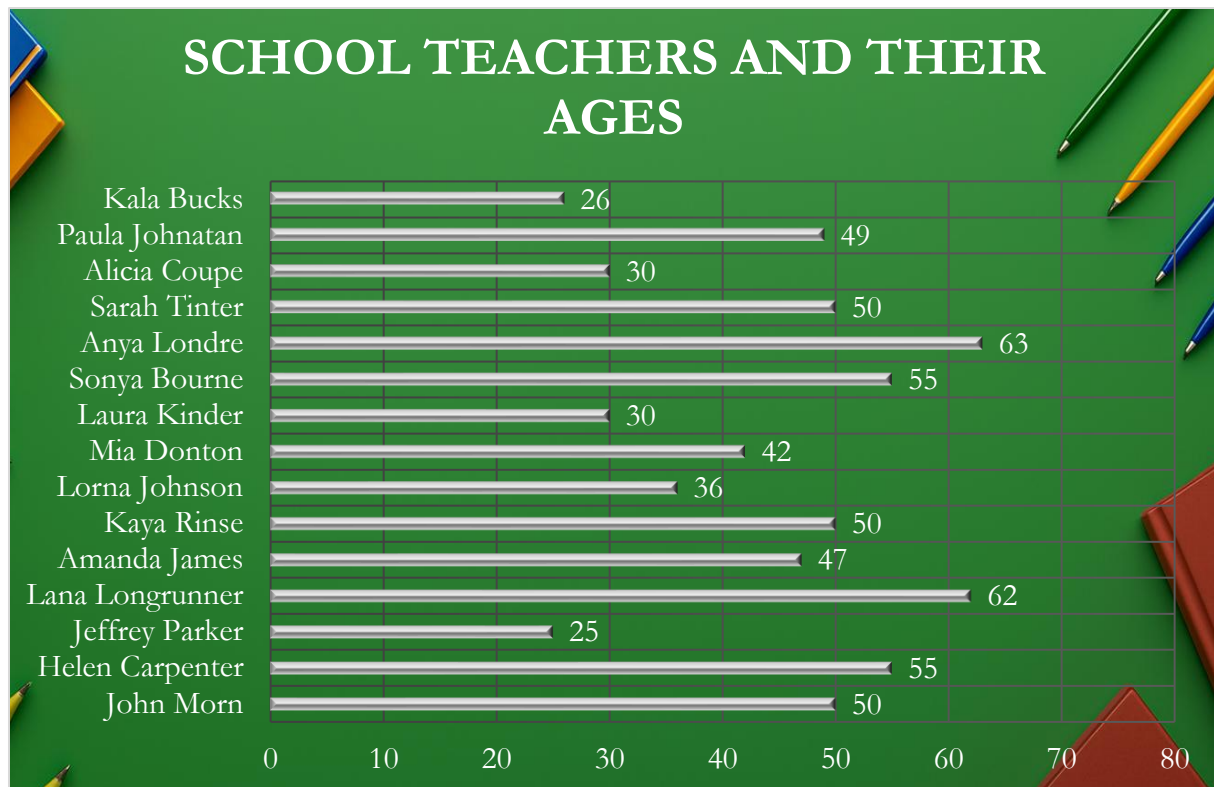
VISUALIZATION 2 – School Teachers

Name	Start Date	End Date	Age
John Morn	2005-09-01 00:00:00	2014-12-15 00:00:00	50
Helen Carpenter	2006-09-01 00:00:00	2017-05-31 00:00:00	55
Jeffrey Parker	2007-09-01 00:00:00	2013-11-29 00:00:00	25
Lana Longrunner	2008-09-01 00:00:00	2020-03-02 00:00:00	62
Amanda James	2009-09-01 00:00:00	2015-05-07 00:00:00	47
Kaya Rinse	2010-09-01 00:00:00	2019-03-18 00:00:00	50
Lorna Johnson	2011-09-01 00:00:00	2021-06-10 00:00:00	36
Mia Donton	2012-09-01 00:00:00	2026-02-28 00:00:00	42
Laura Kinder	2013-09-01 00:00:00	2022-04-18 00:00:00	30
Sonya Bourne	2014-09-01 00:00:00	2022-11-17 00:00:00	55
Anya Londre	2005-09-01 00:00:00	2017-11-28 00:00:00	63
Sarah Tinter	2006-09-01 00:00:00	2015-11-18 00:00:00	50
Alicia Coupe	2007-09-01 00:00:00	2018-01-01 00:00:00	30
Paula Johnatan	2008-09-01 00:00:00	2017-03-17 00:00:00	49
Kala Bucks	2009-09-01 00:00:00	2016-01-15 00:00:00	26

And of course, I am going to make a some changes. Here's the cleaned data:

Name	Start Date	End Date	Age
John Morn	01.09.2005 r.	15.12.2014 r.	50
Helen Carpenter	01.09.2006 r.	31.05.2017 r.	55
Jeffrey Parker	01.09.2007 r.	29.11.2013 r.	25
Lana Longrunner	01.09.2008 r.	02.03.2020 r.	62
Amanda James	01.09.2009 r.	07.05.2015 r.	47
Kaya Rinse	01.09.2010 r.	18.03.2019 r.	50
Lorna Johnson	01.09.2011 r.	10.06.2021 r.	36
Mia Donton	01.09.2012 r.	28.02.2026 r.	42
Laura Kinder	01.09.2013 r.	18.04.2022 r.	30
Sonya Bourne	01.09.2014 r.	17.11.2022 r.	55
Anya Londre	01.09.2005 r.	28.11.2017 r.	63
Sarah Tinter	01.09.2006 r.	18.11.2015 r.	50
Alicia Coupe	01.09.2007 r.	01.01.2018 r.	30
Paula Johnatan	01.09.2008 r.	17.03.2017 r.	49
Kala Bucks	01.09.2009 r.	15.01.2016 r.	26

On the next page, I have the visualization:



(Visualization created by Excel and with the help of AI)

Let me calculate the average age first:

$26+49+30+50+63+55+30+42+36+50+47+62+25+55+50=670 / 15 = \underline{44.67}$ as average age.

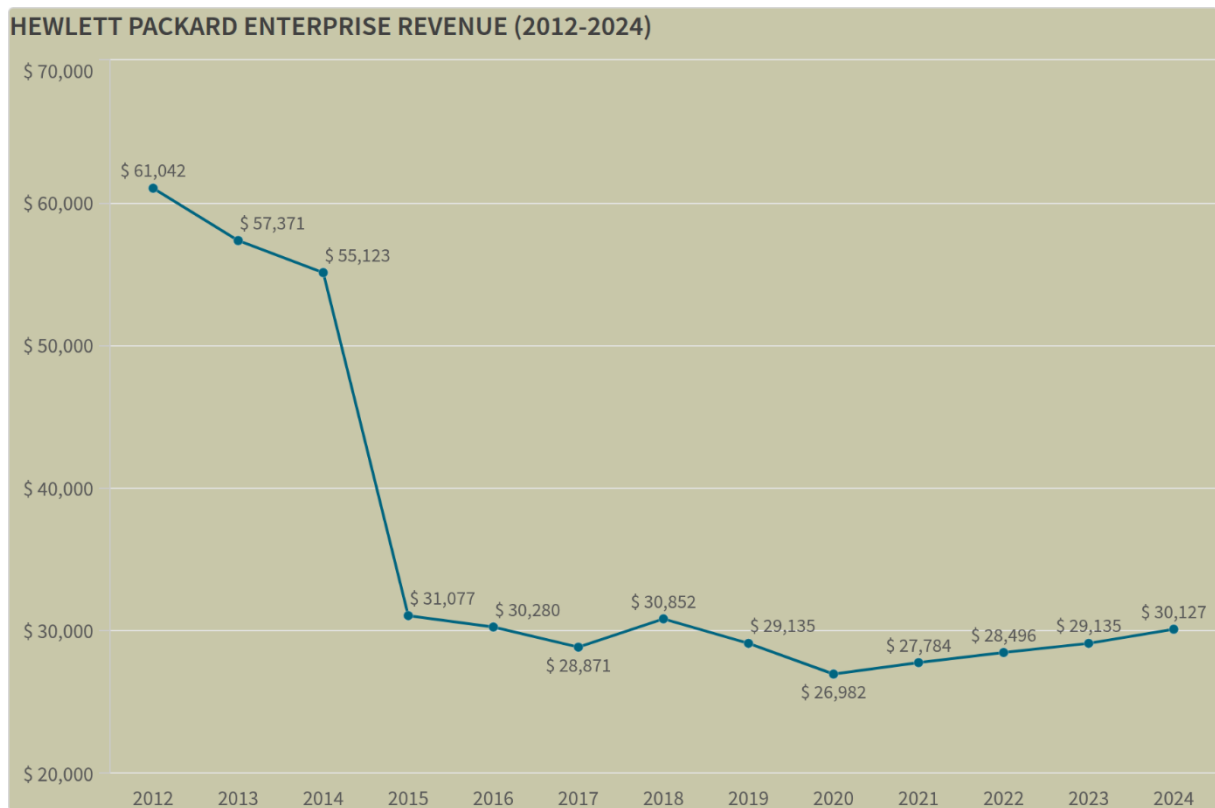
Recommendation: It is always better to make the job more attractive to young people. So, here is what you can do:

- *Increase the salary;*
- *Focus on more interesting study textbooks with more visualization and things to learn you need in the future of your life;*
- *Investing of playgrounds – the children sometimes need to have fun, as people say in countries like Finland, for example. They need to have time for themselves to go out of the learning zone for a moment at least. The teacher and especially the younger ones appreciate that. The young people like sport, so I think that will help;*
- *More attractive training programs for the teachers – about ways to teach children and thing that are needed for their future development.*

VISUALIZATION 3 – Hewlett Packard Enterprise Revenue

year	revenue
2012	\$61,042
2013	\$57,371
2014	\$55,123
2015	\$31,077
2016	\$30,280
2017	\$28,871
2018	\$30,852
2019	\$29,135
2020	\$26,982
2021	\$27,784
2022	\$28,496
2023	\$29,135
2024	\$30,127

Here is data for the annual revenue of Hewlett Packard Enterprise. I am showing you the visualization:



(Visualization made using Qlik Analytics)

Data source:

<https://www.macrotrends.net/stocks/charts/HPE/hewlett-packard-enterprise/revenue>

It looks like that Hewlett Packard Enterprise has a huge profit during the starting period of its creation. And then, after 2014, the revenue are going abruptly down. Possible reason is that there are so many IT companies right now and there is a lot of competition.

Recommendations:

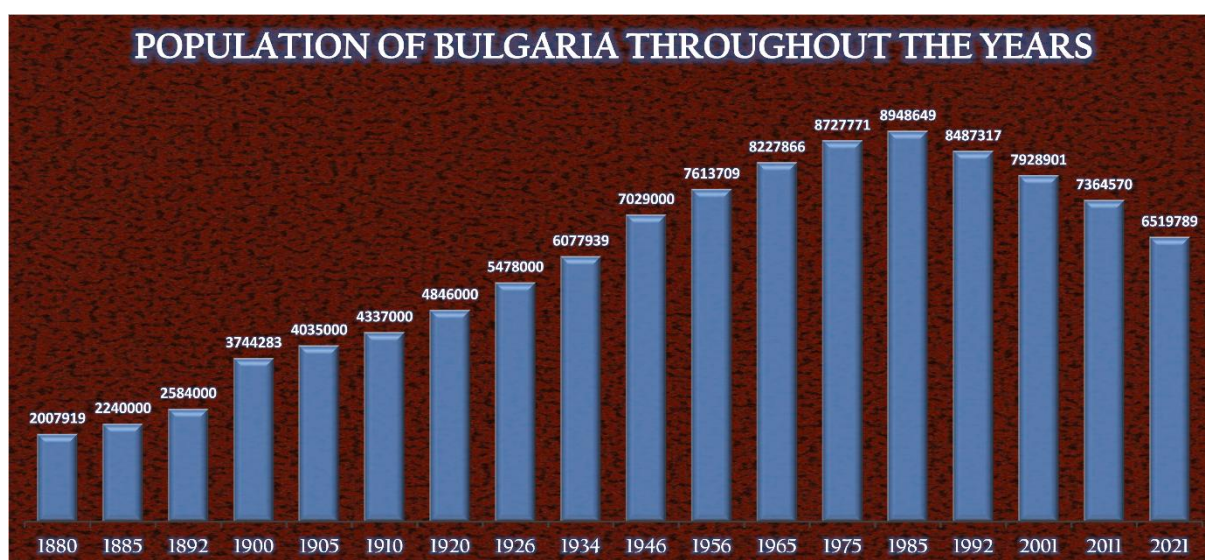
- You can make training programs with high quality for improving the skills and knowledge of the people working and easier to hire new junior people;
- Make entertainment environment for the hardworking people (video games and other similar ways).

VISUALIZATION 4 – THE POPULATION OF BULGARIA

Year	Number
1880	2007919
1885	2240000
1892	2584000
1900	3744283
1905	4035000
1910	4337000
1920	4846000
1926	5478000
1934	6077939
1946	7029000
1956	7613709
1965	8227866
1975	8727771
1985	8948649
1992	8487317
2001	7928901
2011	7364570
2021	6519789

This part is about the people of Bulgaria. How the population is changing during the periods. Is the country going up or going down? We'll see on the chart below.

I am showing you attractive visualization in which you will see the situation.



(Visualization using Excel)

Data Source: Wikipedia – Население на България (in English: Population of Bulgaria)

The population of Bulgaria is going up during the 19th and 20th century. The beginning of the decreasing starts after 1985. According the information I have, that thing has started after 1989 when the socialism in Bulgaria falls and with the appearance of the democracy the reducing begins.

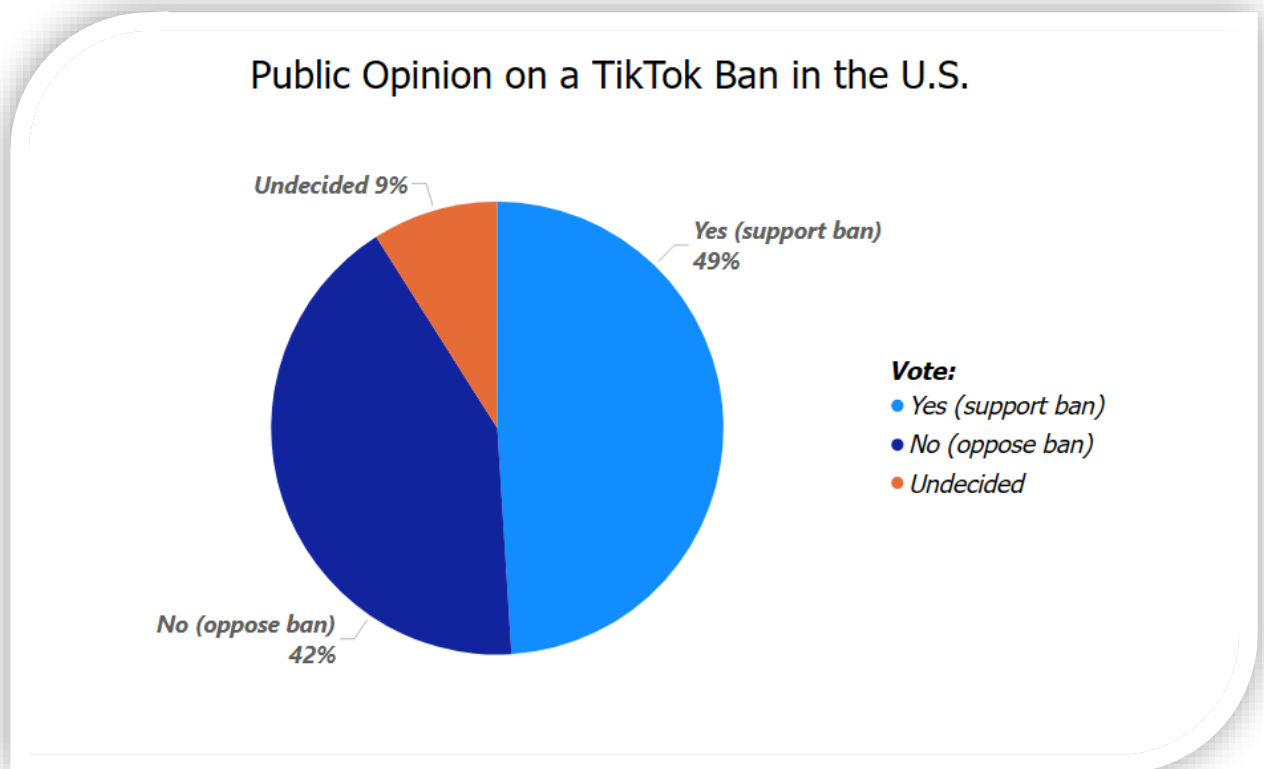
Recommendation: The people in the power of the state have the greatest responsibility for the people in Bulgaria to live well enough. For many years there has been a going down – possible reason is the bad control in the country. The national assembly is the one that must take care of the problems.

VISUALIZATION 5 – TIKTOK BAN IN THE USA

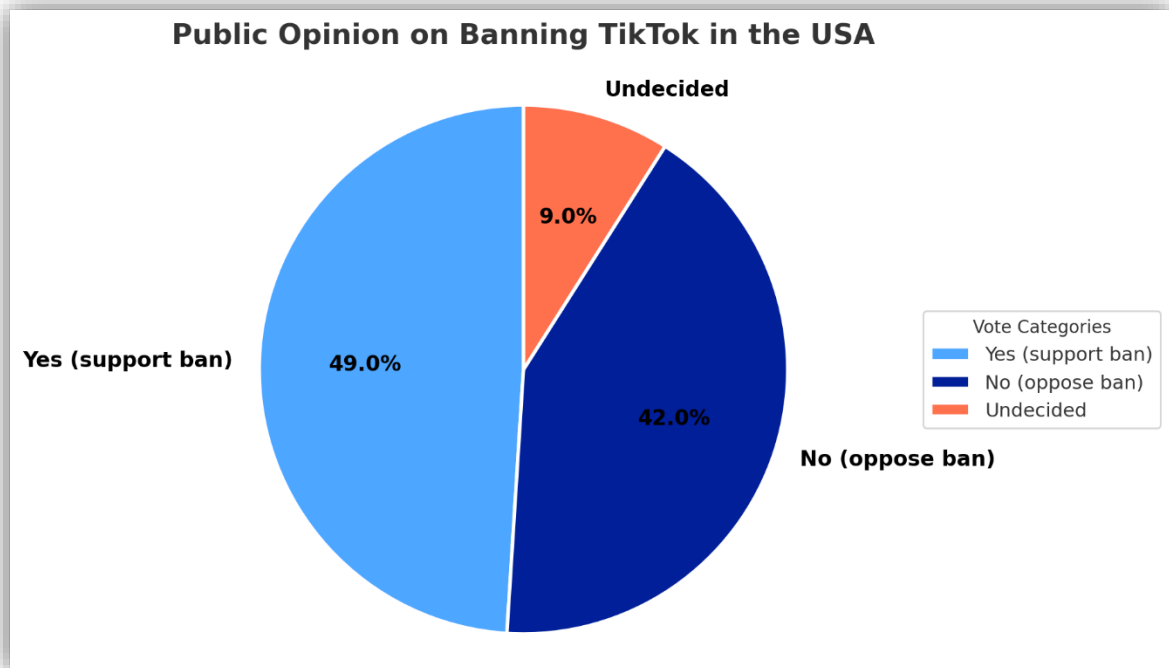
Response	Percentage
<i>Yes (support ban)</i>	<i>49%</i>
<i>No (oppose ban)</i>	<i>42%</i>
<i>Undecided</i>	<i>9%</i>

This is a voice for banning TikTok in the United States of America. The people's opinion about it – should the platform be stopped? Or it isn't a problem and they can use it independently?

I am going to show you a visualization, made with the help of another program, different than Microsoft Excel:



(Visualization crated using Power BI)



(This is the same picture, but improved by AI)

As you can see, even the AI is a big helper sometimes, makes the work easier than we thought.

Recommendation: TikTok is a platform for sharing, for publicity. The problem is that there is a lack of enough control. Even the children are allowed to upload. It should be actually for educational purpose, because that is the key to survive in the world we live.

I think it is better for TikTok to remain but with enough rules and control.

THIS REPORT WAS CREATED BY:



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Programs and platforms used for this project:

Microsoft Word;

Microsoft Excel;

Microsoft Power BI;

Qlik Analytics;

AI